VILNIUS GEDIMINAS TECHNICAL UNIVERSITY

FACULTY OF FUNDAMENTAL SCIENCES
DEPARTMENT OF INFORMATION SYSTEMS

Application of Artificial Intelligence in Information Security Technologies

Practical Work 2

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1 intro

For this lab, a new dataset was chosen, because the ember2018 dataset which comes with scripts for preprocessing, is deprecated. The scripts are necessary for vectorization of the dataset, but without them we were not able to process the dataset which contains nested JSON Structures.
The newly chosen dataset is the "Malware static and dynamic features VxHeaven and Virus Total" retrieved from the machine learning repository of the UC Irvine. (“Malware static and dynamic features VxHeaven and Virus Total”, 2019)

For this work the following tasks, according to the instructions are perfomed:

1. Reload the clean PW1 dataset and splits.
2. Plot(s): label distribution bar plot/ pie chart
3. In the context of your experiment,
    - What does your imbalance imply for precision/recall trade-off?
    - How might imbalance affect model training stability?
4. Choose the segmentation method appropriate to your domain
5. Apply at least 3 balancing methods that match your dataset type on the training split only

2 Tasks

2.1 Load and clean dataset

Due to the fact that the dataset in use was switched since the previous lab, for this tasks the cleaning steps are performed again on the new dataset.
Steps in the cleaning pipeline:

- Drop duplicate rows
- Separate label column
- Coerce features to numeric (non-numeric $\rightarrow$ NaN)
- Impute missing with 0 (common for sparse indicator features)
- Drop constant (zero-variance) features
- Keep a record of cleaned feature columns

```
Raw shape: (6248, 1089)
Raw label counts:
 label
1    5653
0     595
```

Listing 1: outputs of the cleaning steps

2.2 Plot label distribution

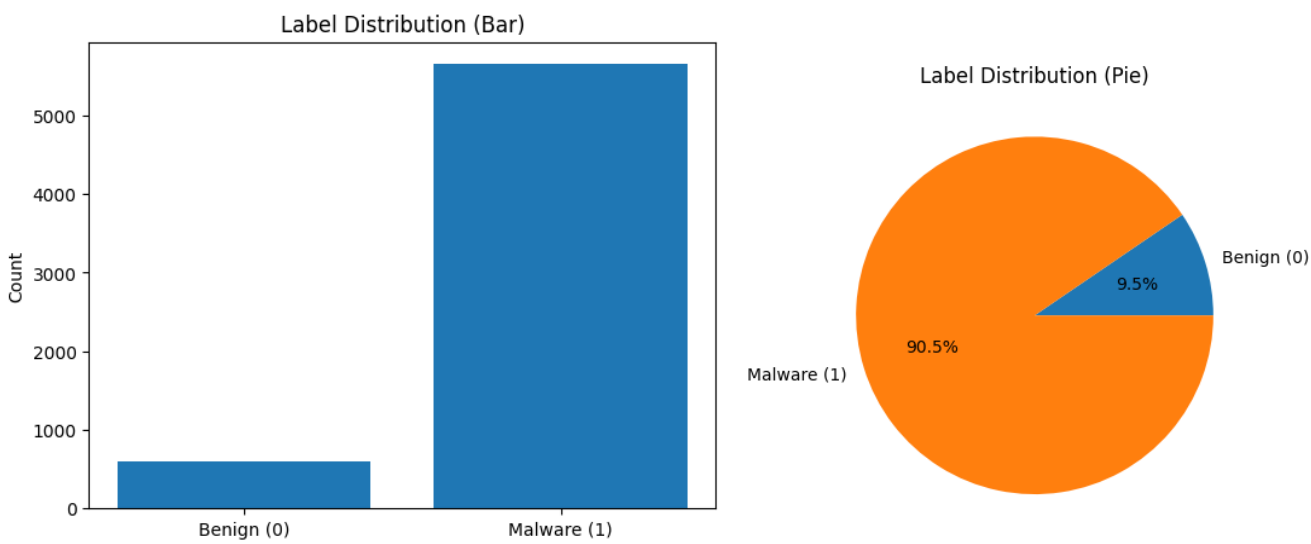

Figure 1: plots of the dataset label distribution

2.3 Investigate label imbalance

What does your imbalance imply for precision/recall trade-off?

explanation 123

How might imbalance affect model training stability?

explanation 123

2.4 Data segmentation

2.5 Data balancing

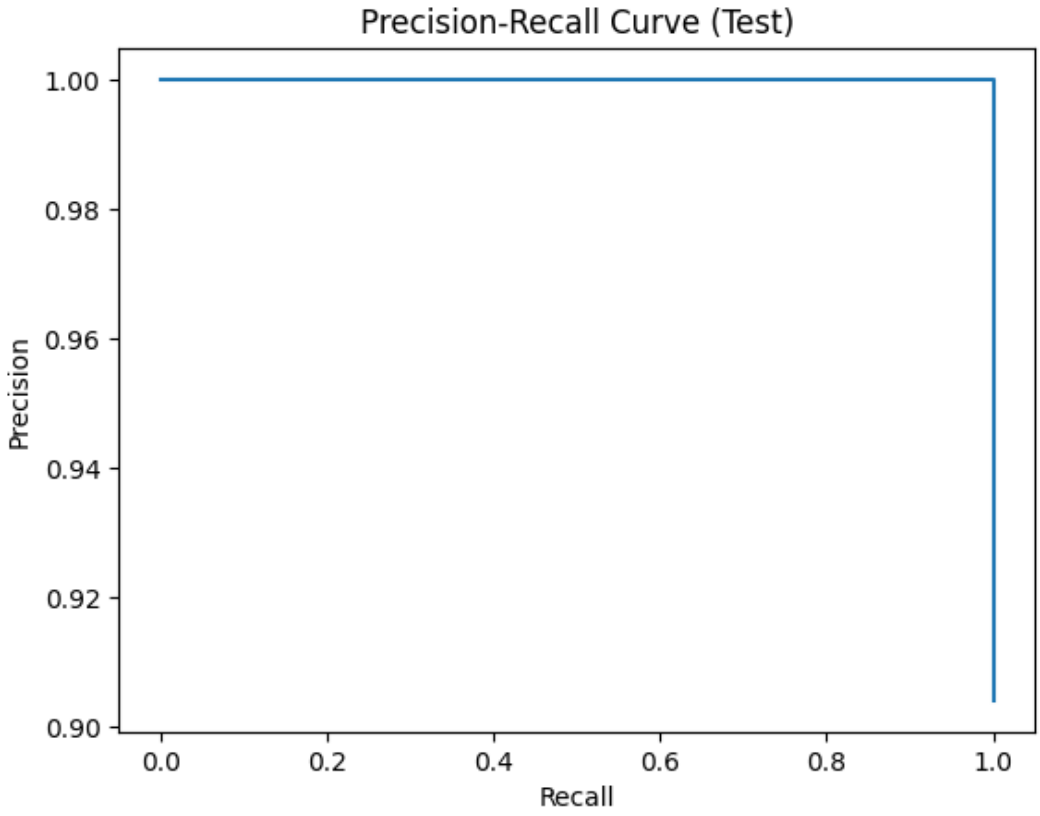

Figure 2: PR curve for XGBoost model

3 Results

4 conclusion

create sections and subsections with \section{title} or \subsection:

5 section

5.1 subsection

with \begin{itemize}you create bullet points

- item 1
- item 2

with \begin{enumerate} you create numbered items

1. first intem
2. second item

insert images like the following (see source file):

Figure 3: caption

you can also use \vspace to create a space

like this. you can use double backslash or an empty line to go to the next line
next line
with \newpage you go to the next page.

this references fig. 3
you can also reference listing 2 `inline command`

```
text shown in the console
like this
```

Listing 2: caption

\textbf makes text bold and *textit makes text italic*
To cite something you put the citation into the refs.bib file in the bibtex format (can be found on all major publisher sites). The you can cite with parenthesis with \parencite{abbreviation on top of the bibtex item}: (Rana et al., 2022) or without parenthesis with \cite: Rana et al., 2022

References

Malware static and dynamic features VxHeaven and Virus Total [DOI: https://doi.org/10.24432/C58K6H]. (2019).

Rana, M. S., Nobi, M. N., Murali, B., & Sung, A. H. (2022). Deepfake detection: A systematic literature review. *IEEE access*, *10*, 25494–25513.